12TH BIOLOGY · CHAPTER 20 · FULL CHAPTER NOTES

Biotechnology

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION AND 20.1 GENETIC ENGINEERING

Q.1 [Short] Define biotechnology. What is its main focus?

Biotechnology means “technology based on biology”. It is the use of organisms, biological processes or systems to make products that improve human lives and the environment. Its main focus is to solve complex challenges in medicine, agriculture and industry.

Q.2 [Short] Differentiate between traditional and modern biotechnology.

Traditional biotechnology has been used for thousands of years in the form of fermentation, to make food products like bread, cheese and beer. **Modern biotechnology** emerged in the 20th century, after the discovery of the structure of DNA and of advanced molecular techniques.

Q.3 [Short] What is genetic engineering? Name its two basic technologies.

Genetic engineering is the alteration of an organism’s DNA to change its characters (traits) or capabilities. It involves inserting, deleting or modifying genes, and sometimes transferring genes between different species. It is the true “engine” of modern biotechnology. It relies on two basic technologies:

1. **Recombinant DNA technology:** physically cutting a piece of DNA from one organism and “pasting” it into the DNA of another.
2. **Polymerase chain reaction (PCR):** rapidly amplifying a tiny segment of DNA into a large quantity.

20.2 RECOMBINANT DNA TECHNOLOGY

Q.4 [Short] What is recombinant DNA technology?

Recombinant DNA technology means cutting, splicing and combining DNA molecules from different sources into a single hybrid molecule. It also involves introducing the recombinant DNA (rDNA) into a host (e.g., bacteria) to produce new genetic combinations or specific proteins.

Q.5 [Short] Name the tools used in recombinant DNA technology.

(1) Gene of interest, (2) restriction endonucleases, (3) vectors, (4) DNA ligase and (5) expression system.

Q.6 [Short] What is the gene of interest? How can it be obtained?

The gene of interest is the gene which is to be manipulated. It can be obtained in three ways:

1. Direct synthesis of the gene in the lab by using enzymes such as polymerases and ligases.
2. Reverse transcription of its mRNA by using reverse transcriptase enzymes.
3. Cleavage from the chromosome by using restriction endonucleases.

Q.7 [Long] Write a detailed note on restriction endonucleases.

Restriction endonucleases are enzymes that cut specific sequences of nucleotides in DNA. They are also called **molecular scissors**.

- There are many kinds of restriction endonucleases. Each kind recognizes a specific nucleotide sequence in DNA, called the **restriction site** or recognition region.
- The enzyme binds to the DNA at that sequence and cuts it at a particular place within the site.
- Restriction sites have **palindromic sequences** of four to eight nucleotides: the two strands have the same nucleotide sequence running in opposite directions.
- The enzyme cleaves both strands. After cleavage, both DNA fragments have single-stranded ends, a few nucleotides long, called **sticky ends**; they are complementary to each other.
- **Value:** any two DNA fragments produced by the same restriction endonuclease can be joined together. So the source DNA and the vector are cut with the same enzyme.

Q.8 [Short] What is a restriction site? What is meant by a palindromic sequence?

The specific nucleotide sequence recognized by a restriction endonuclease is called the restriction site or recognition region. It is palindromic: the nucleotides at one end are complementary to those at the other end, so both strands have the same sequence running in opposite directions. It is four to eight nucleotides long.

Q.9 [Short] What are sticky ends? Why are they useful?

After cleavage by a restriction endonuclease, both DNA fragments have single-stranded ends a few nucleotides long. These ends are complementary to each other, so they are called sticky ends. Because of them, any two DNA fragments cut by the same enzyme can be joined together.

Q.10 [Long] What are vectors? Describe the different kinds of vectors.

Vectors (molecular carriers) are special molecules used to introduce the gene of interest into host cells. The gene of interest is attached to the vector, which becomes a recombinant DNA (rDNA); the rDNA is then introduced into the host cells. Bacterial **plasmids** are the most commonly used vectors. Other vectors are:

- Bacteriophages (e.g., phage lambda, bacteriophage P1)
- Cosmids (plasmids that contain a small region of bacteriophage DNA called the cos sequence)
- Bacterial artificial chromosomes (BACs)
- Yeast artificial chromosomes (YACs)
- P1 artificial chromosomes (features of both P1 vectors and BACs)

- Animal viruses (e.g., human cold virus), which carry genes into monkey and human cells

Q.11 [Short] What are cosmids?

Cosmids are plasmids that contain a small region of bacteriophage DNA called the cos sequence. They are used as vectors.

Q.12 [Short] Why is DNA ligase called molecular glue?

DNA ligase joins two DNA fragments, so it is called **molecular glue**. It forms the phosphodiester linkage between two adjacent nucleotides.

Q.13 [Short] What is an expression system? Why are bacteria ideal for it?

An **expression system** is a suitable organism that can act as host for the rDNA and can express the gene of interest. Its selection depends on the type of vector used. Bacterial cells are an ideal expression system because of their short generation time and simple genetic system.

Q.14 [Long] Describe the steps of recombinant DNA technology.

1. Formation of recombinant vector (plasmid)

DNA cleavage: The source DNA is cleaved with a restriction endonuclease and the gene of interest is isolated. The DNA of the vector (e.g., plasmid) is cleaved with the same endonuclease. So compatible sticky ends are produced on both.

Production of recombinant DNA: The cleaved vector and the gene of interest are incubated together in the presence of DNA ligase. It joins them by phosphodiester linkages, and the vector becomes a recombinant vector.

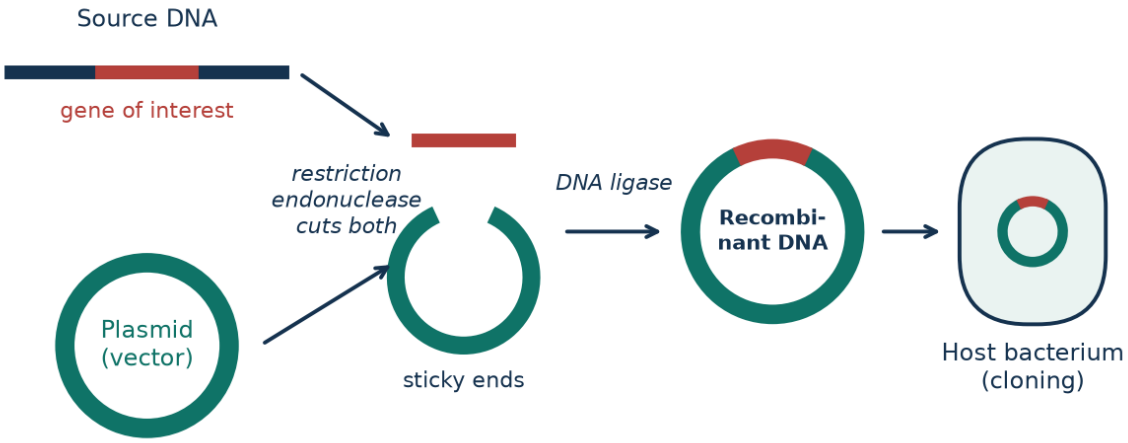
2. Cloning of recombinant vector

The recombinant DNA and the expression system (usually bacteria) are incubated in the same medium. The bacterial cells take up the recombinant DNA. The bacteria multiply into clones, which also contain identical copies (clones) of the gene of interest.

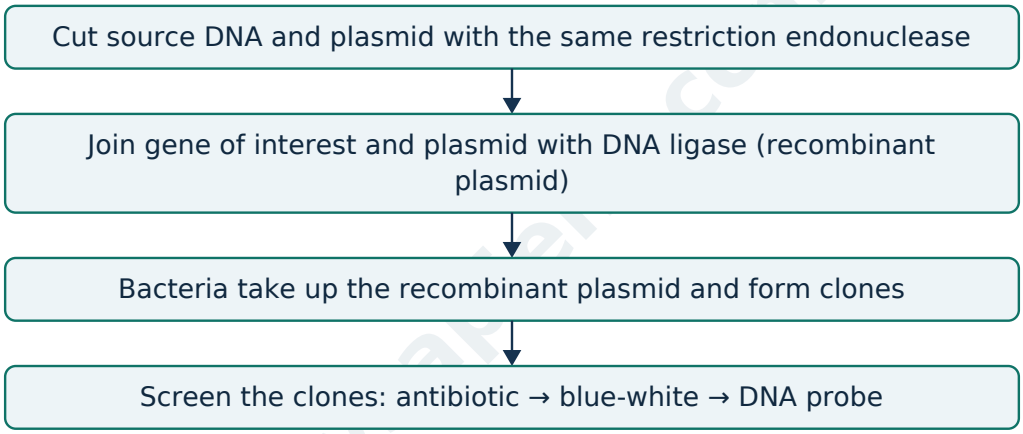
3. Screening of clone library

1. **Selecting for the vector (antibiotic resistance):** The plasmid contains an antibiotic resistance gene. The bacteria are grown on a medium containing that antibiotic (e.g., ampicillin). Only the cells that took up a plasmid survive and grow.
2. **Selecting for the gene of interest (blue-white screening):** The lacZ′ gene of the plasmid makes an enzyme which metabolizes a specific sugar and turns it blue. In a non-recombinant plasmid, lacZ′ stays functional and the colony is blue. An inserted gene of interest “breaks” lacZ′, so the desired recombinant clones remain **white**.
3. **Identifying the specific gene (DNA hybridization):** A radioactive probe (a DNA strand complementary to the gene of interest) is used. It hybridizes only to colonies containing the gene. Auto-radiography (photographic film) locates these colonies on the master plate.

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Formation and cloning of recombinant DNA



Steps of recombinant DNA technology

Q.15 [Short] Why are the source DNA and the vector cut with the same restriction endonuclease?

The same enzyme cuts the same restriction site in both DNAs. So the gene of interest and the cleaved vector get compatible (complementary) sticky ends, which can pair, and DNA ligase can join them.

Q.16 [Short] How are bacteria without a plasmid eliminated during screening?

The plasmid contains an antibiotic resistance gene. The bacteria are grown on a medium containing that antibiotic (e.g., ampicillin). Cells without a plasmid die; only cells that took up a plasmid survive and grow.

Q.17 [Short] What is blue-white screening? Why do recombinant clones remain white?

It distinguishes a plain plasmid from a recombinant plasmid. The lacZ′ gene of the plasmid makes an enzyme that turns a specific sugar blue. In a recombinant plasmid, the inserted gene of interest “breaks” the lacZ′ gene, so no blue colour is formed and the colony remains white. Blue colonies have plasmids without the gene.

Q.18 [Short] What is a probe? How is it used in DNA hybridization?

A probe is a radioactive DNA strand complementary to the gene of interest. It sticks (hybridizes) only to the colonies that contain the gene of interest. Auto-radiography on photographic film then shows the position of these colonies on the master plate.

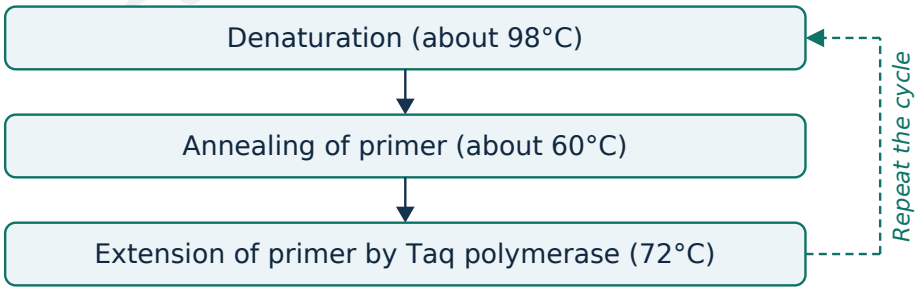
20.3 POLYMERASE CHAIN REACTION (PCR)

Q.19 [Long] Explain the polymerase chain reaction (PCR) and its steps.

In the **polymerase chain reaction (PCR)**, multiple identical copies of the gene of interest are made. The enzyme DNA polymerase polymerizes a given piece of DNA again and again. Kary Mullis developed PCR in 1983 and won the Nobel Prize in Chemistry in 1993.

Steps of PCR

1. **Denaturation:** An excessive amount of primer (a sequence of 20 to 30 nucleotides) is mixed with the DNA fragment. The mixture is heated to about 98°C. The double-stranded DNA breaks into single strands, each of which can act as a template.
2. **Annealing of primer:** The solution is cooled to about 60°C. The forward and backward primers anneal (hybridize) to the single-stranded DNA at their complementary regions.
3. **Extension of primer:** A very heat-stable DNA polymerase, **Taq polymerase**, and all four types of nucleotides are added. The enzyme adds new nucleotides to the 3′ end of the primer and copies the rest of the fragment (optimum temperature 72°C). Now there are two copies of the original DNA fragment.
4. **Repeating the cycle:** Steps 1 to 3 are repeated and the two copies become four. No more polymerase is needed, because heating does not harm this enzyme. Each cycle takes one or two minutes and doubles the number of DNA molecules. After 20 cycles, a single fragment gives more than one million copies.



One cycle of PCR

Q.20 [Short] Why is the technique called polymerase chain reaction? Who developed it?

The enzyme DNA polymerase polymerizes a given piece of DNA again and again, in a chain of cycles, so that multiple copies are produced. Kary Mullis developed it in 1983 and won the Nobel Prize in Chemistry in 1993.

Q.21 [Short] What is Taq polymerase? Why is it used in PCR?

Taq polymerase is a temperature-tolerant DNA polymerase isolated from *Thermus aquaticus*, a bacterium found in hot springs. It is stable and active at near-boiling temperatures. So the heating in each cycle does not harm it, and no fresh enzyme has to be added.

Q.22 [Short] State the temperatures used in the three steps of PCR.

Step	Temperature	Event
Denaturation	About 98°C	Double-stranded DNA separates into single strands
Annealing	About 60°C	Primers bind to complementary regions
Extension	72°C (optimum)	Taq polymerase adds nucleotides to the 3′ end of the primer

Q.23 [Short] How many copies of DNA are produced after 20 cycles of PCR?

Each cycle doubles the number of DNA molecules (1 → 2 → 4 → 8 …). After 20 cycles, a single fragment gives 2^{20}, i.e., more than one million copies. Each cycle takes only one or two minutes.

Q.24 [Short] What is a genomic library?

It is the collection of the total DNA from a single organism. PCR is used to build it. Genomic libraries are commonly used for sequencing DNA.

Q.25 [Long] Describe the applications of PCR.

1. It is an efficient diagnostic technique for detecting the genotypes of infectious agents.
2. It detects genetic mutations responsible for certain genetic diseases and cancers.
3. In criminal investigations, “DNA fingerprints” are prepared from a tiny sample of dried blood or from the base of a single hair.
4. Physicians detect genetic defects in very early embryos by amplifying the DNA of a few sloughed-off cells.
5. It is used to examine the DNA of extinct species, if even a minuscule amount of their DNA remains intact.
6. It is also used to build a genomic library (the collection of the total DNA of a single organism).

20.4 GENETICALLY MODIFIED ORGANISM (GMO)

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Q.26 [Short] Define genetically modified organism. What are transgenic organisms?

A **genetically modified organism (GMO)** is a plant, animal or microbe in which the genome has been changed. **Transgenic organisms** are a specific subset of GMOs that have received DNA from a different, unrelated species. About 90% of the canola, cotton, corn, soybean and sugar beet grown in the world are transgenic.

Q.27 [Long] Describe the methods of gene transfer into a target cell.

Method	How the gene enters the target cell
Electroporation	Electric current creates pores in the cell wall or membrane; the gene enters; the pores close when the current is discharged.
Microinjection	A fine needle injects the gene directly into the nucleus.
Biolistic transformation	The gene, coated with heavy metal ions, is shot into the cell by mechanical force, e.g., gene gun.
Protoplast mediated technique	The cell wall and cell membrane are removed; the protoplast is mixed with a medium containing the gene.
Chemical methods	Naked DNA is fused with a chemical agent; the complex enters the cytoplasm by diffusion or endocytosis.
Virus mediated gene transfer	Adenovirus, adeno-associated virus, HIV, lentivirus and other retroviruses carry the gene.

Q.28 [Short] How is a whole multicellular GMO produced?

The gene is transferred into a gamete or into gamete-producing tissue, through the cell membrane or cell wall, without destroying the cell. Such gametes are then fused to produce the GMO.

Q.29 [Long] Describe the uses of transgenic bacteria.

Genes from a wide range of organisms can be added to a plasmid and inserted into bacteria. Transgenic bacteria synthesize large amounts of the proteins encoded by the foreign genes. They are used to:

1. **Pharmaceutical products:** insulin (diabetes), clotting factors (haemophilia), hepatitis B vaccine, Covid-19 vaccine, human growth hormone (dwarfism), interferon (some cancers), erythropoietin (anaemia) and tissue plasminogen activator (dissolves blood clots).
2. **Food products:** alpha-amylase (converts starch to simple sugars), chymosin (clots milk protein for cheese making) and pectin-esterase (improves fruit juice clarity).
3. **Enzymes that degrade toxins**, converting pollutants into a less toxic form.
4. **Environment and mining:** leaching copper from ore, cleaning up mercury pollution and detecting arsenic in drinking water.

Q.30 [Long] What are transgenic animals? Give examples of their uses.

A transgenic animal carries a foreign gene to produce the desired protein. Transgenic animals are used for producing drugs, enhancing yields, increasing resistance to disease etc.

1. Transgenic dairy cows have human genes and produce milk that is the same as human breast milk.
2. Transgenic goats produce milk with strong spiderweb-like silk proteins.
3. Transgenic goats also produce the human enzyme lysozyme in milk, to fight diarrhoea-causing bacteria.
4. Transgenic fish show better growth, reduced food intake, increased cold tolerance and disease resistance.
5. In gene therapy, genetically modified viruses deliver genes to humans to cure inheritable diseases, e.g., abnormal development of T and B cells and inherited blindness.
6. The first human biological drug from a transgenic goat was an anticoagulant, used to reduce blood clots during surgery or childbirth.
7. Malaria-resistant mosquitoes carry a gene that reduces the development of *Plasmodium* in them.

Q.31 [Short] Why are mice used as a model for investigating diseases?

Transgenic mice have been a key model since the mid-1980s. The genome sequence of the mouse has been completely analysed and it is very similar to the human genome.

Q.32 [Short] Why does most gene-insertion research in humans focus on somatic cells?

Foreign genes may be inserted in germline cells or in somatic cells. Insertion in germline cells gives inheritable changes, which carries risks and ethical issues. So most research has focused on somatic cells.

Q.33 [Long] Describe the main purposes of producing transgenic plants.

Many plant cells are **totipotent**, i.e., a single cell can form a new plant. So biologists grow a whole new plant from one transgenic cell. The main purposes are:

1. **Insect resistance:** genes of *Bacillus thuringiensis* (a soil bacterium) have been inserted in cotton, rice, maize, potato, tomato, brinjal, cauliflower, cabbage etc. They make proteins that kill insect pests but do not harm other useful insects.
2. **Herbicide resistance:** many crops have been made resistant to glyphosate (a herbicide).
3. **Better nutrition (golden rice):** genes from beans, *Aspergillus* fungus and wild rice increase the iron content; genes from daffodil make beta-carotene (a precursor of vitamin A).
4. **Biopharmaceuticals:** GMO plants and plant cells produce cytokines, hormones, antibodies, enzymes and vaccines.

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Q.34 [Short] What is golden rice? Why was it made?

Golden rice is a transgenic rice. Genes from beans, *Aspergillus* fungus and wild rice were inserted to increase its iron content, and genes from daffodil to make beta-carotene (a precursor of vitamin A). It is grown and eaten in areas with a shortage of dietary iron and vitamin A. Each year iron deficiency affects 1.4 billion women, and vitamin A deficiency kills 670,000 children and causes 500,000 cases of childhood blindness.

Q.35 [Short] How do Bt crops resist insects?

They carry genes of the soil bacterium *Bacillus thuringiensis*. These genes make proteins that kill insect pests but do not harm other useful insects.

20.5 FORMATION OF HUMAN INSULIN IN BACTERIA

Q.36 [Long] Explain the formation of human insulin protein in bacteria.

The production of human insulin was the first major triumph of genetic engineers. The two polypeptide chains are produced separately in bacteria and then joined chemically.

1. **Preparation of the two genes:** Two different DNA sequences are synthesised, one for Chain A and one for Chain B.
2. **Construction of recombinant plasmids:** The DNA for Chain A is inserted into one set of plasmids and the DNA for Chain B into another. The insulin gene is often attached next to a bacterial gene (like lacZ), which helps the bacteria to recognize and express the foreign DNA.
3. **Transformation of host cells:** The recombinant plasmids are introduced into two separate cultures of *Escherichia coli*. Culture 1 is the factory for Chain A and culture 2 for Chain B.
4. **Fermentation and protein expression:** The bacteria are grown in large industrial bioreactors. As they multiply, they read the human DNA and produce large quantities of the insulin chains.
5. **Extraction and chemical joining:** The bacteria are harvested and the chains are extracted and purified. Chain A and Chain B are mixed and linked by di-sulfide bonds between specific amino acids. This gives functional human insulin, identical to that of a healthy pancreas.

Q.37 [Short] Why are two separate bacterial cultures used for making insulin?

Insulin consists of two distinct polypeptide chains, A and B. A separate DNA sequence is made for each chain and inserted into a separate set of plasmids. So one culture of *E. coli* makes Chain A and the other makes Chain B.

Q.38 [Short] How are the two chains of insulin joined?

The purified Chain A and Chain B are mixed in the laboratory. They are linked by di-sulfide bonds (sulfur-to-sulfur bridges) between specific amino acids. This gives the functional, three-dimensional insulin molecule.

Q.39 [Short] **Why is the insulin gene attached next to a bacterial gene like lacZ?**

It helps the bacteria to “recognize” and express the foreign (human) DNA.

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